

12. 增广矩阵:

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -4 & 6 & -1 & 7 \end{bmatrix}$$

$$\downarrow \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ -4 & 6 & -1 & 7 \end{bmatrix}$$

$$\downarrow \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & -6 & 15 & -9 \end{bmatrix}$$

$$\downarrow \begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 1 & -\frac{5}{2} & 2 \\ 0 & -6 & 15 & -9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 1 & -\frac{5}{2} & 2 \\ 0 & 0 & 0 & 3 \end{bmatrix} \text{ 无解.}$$

增广矩阵如:

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ -1 & 1 & 5 & 2 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & -2 & 5 & 7 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 1 & 1 & 0 \end{array} \right]$$

-3倍加到第3行

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 0 & \frac{7}{2} & \frac{7}{2} \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 0 & 1 & 1 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & 1 & -\frac{5}{2} & -\frac{7}{2} \\ 0 & 0 & 1 & 1 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$\therefore \begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 1 \end{cases}$$

$$\begin{matrix} -3 & 9 & -12 \\ & 4 & \\ & -\frac{5}{2}x_3 & = -14 \end{matrix}$$